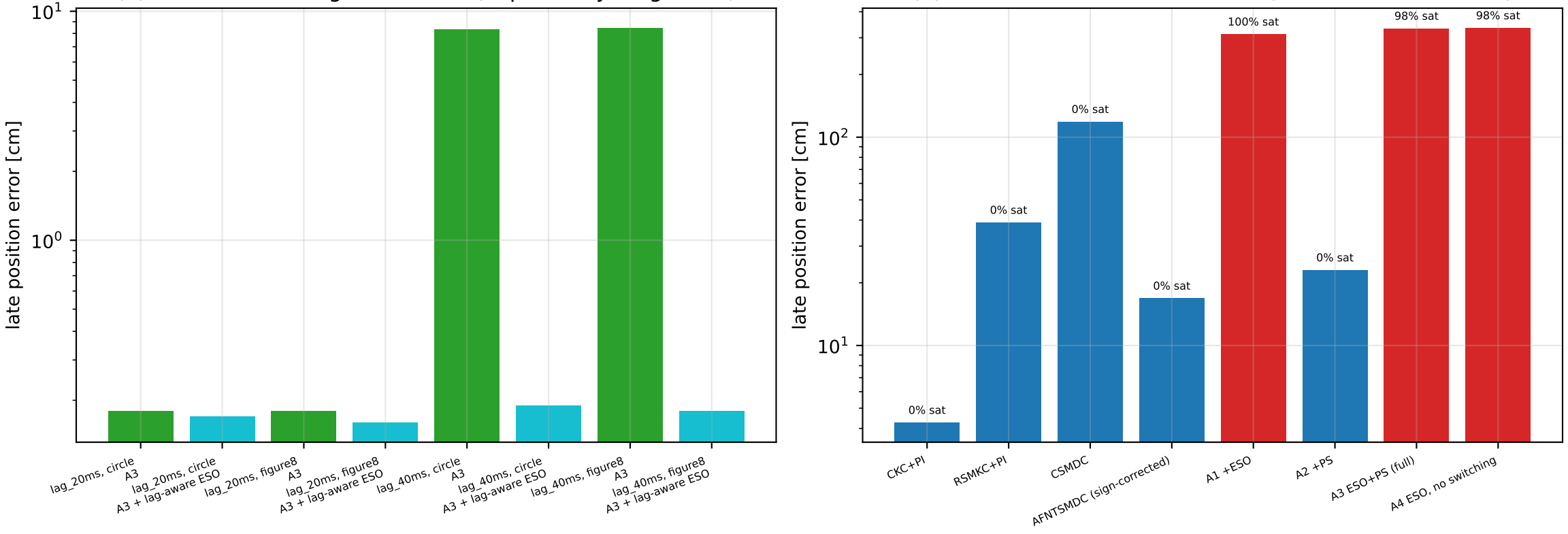F7: limitations of the improved design (exploratory diagnostics)